Federated Bandits with a Server

Near-optimal regret with little communication

CS6007 project notes

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What this folder is about. M agents play the *same* bandit and talk only through a central server. The question is how little communication still gives the regret of one agent with all the data. These papers define the regret–communication trade-off for linear and multi-armed bandits that our server results mirror.

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1 Wang, Hu, Chen & Wang (2019/ICLR 2020): distributed bandit learning

Y. Wang, J. Hu, X. Chen, L. Wang. Distributed Bandit Learning: Near-Optimal Regret with Efficient Communication. ICLR 2020 (arXiv:1904.06309).
File: `1904.06309v2.pdf`

Setting.

- M agents, one server; communication measured as the total amount of transmitted data.

What they did.

- **DEMAB** for multi-armed bandits (distributed elimination).
- **DELB** for linear bandits with a fixed action set (phased elimination with optimal design).
- **DisLinUCB** for linear bandits with time-varying action sets: sync when a log-determinant ratio of the Gram matrix crosses a threshold (event-triggered).

Main results.

- DEMAB: near-optimal regret with $O(M \log(MK))$ communication, independent of T, matching a lower bound up to a log factor.
- DELB: regret $O(d\sqrt{TM}\log)$ with communication $O((Md + d\log\log d)\log T)$.
- DisLinUCB: regret $O(d\sqrt{MT}\log^2 T)$ with communication $O(M^{1.5}d^3)$.

Why it matters for our project.

- Our event-triggered sync is the per-bin analogue of DisLinUCB's determinant trigger; our Theorem 12 mirrors their counting argument.

Open problems.

- **Our reading:** DisLinUCB's communication is worse in M and d than DELB's; closing that gap for time-varying action sets was left open (addressed by later work).
- **Our reading:** Communication is counted in transmitted numbers, not bits (Salgia & Zhao, below, counts bits).

2 Salgia & Zhao (2022): distributed linear bandits under communication constraints

S. Salgia, Q. Zhao. Distributed Linear Bandits under Communication Constraints. arXiv:2211.02212, 2022.
File: `2211.02212v1.pdf`

Setting.

- Distributed linear bandits with a server; uplink and downlink channels have fixed capacity (bits per use).

What they did.

- Information-theoretic **lower bounds** on the bits needed for sublinear regret.
- **PLS** (Progressive Learning and Sharing): learn and share the unknown vector one bit at a time.
- A variant for sparse linear bandits.

Main results.

- PLS is order-optimal in regret using $O(d\log T)$ uplink bits and $O(d(\log M + \log T))$ downlink bits.
- These communication costs match the lower bounds; sparsity improves the trade-off further.

Why it matters for our project.

- The model for a *communication lower bound*, which we identified as the theory missing from our server results (our sync bound is 60–75× loose).

Open problems.

- **Our reading:** Only a server setting; bit-level lower bounds over graphs are open.
- **Our reading:** Linear rewards only; nonlinear or unknown-link versions are open.

3 Open problems across this folder

- **Matching communication lower bounds** exist for linear bandits (bits) and MAB (messages), but not for nonparametric or single-index bandits.
- Event-triggered sync is analysed in the worst case; instance-dependent communication bounds are not.
- All agents are identical here; folder 05 covers what changes when they are not.